

FOUNDATION OF COMPUTER THEORY - HOMEWORK 02

Q.1 Consider the following statements:

a) For all languages A and B , if $A \subseteq B$ and A is regular, then B is regular.

Soln FALSE

Counterexample:

- Let $B = \{a^i b^j \mid i > 0\}$ and
 $A = \{ab\}$

- Now A is regular $A \subseteq B$, A is regular but B is not.

Q.1 b) For all languages A and B , if $A \subseteq B$ and B is regular, then A is regular.

Soln FALSE

- Let $B = \Sigma^*$ where $\Sigma = \{a, b\}$ and
 $A = \{a^i b^j \mid i > 0\}$

- Now $A \subseteq B$, B is regular but A is not.

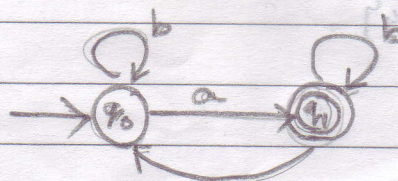
Q.2. Apply the cartesian product construction from class to the following language:

$L = \{w \in \{a, b\}^* \mid w \text{ has odd number of } a\text{'s and contains the substring } bb\}$

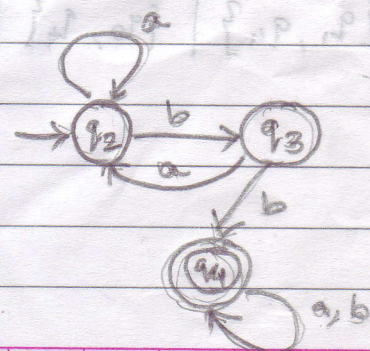
Soln

- Let $L(M) = L(M_1) \cup L(M_2)$
 where $L(M_1) = \{w \in \{a, b\}^* \mid w \text{ has odd number of } a\text{'s}\}$
 $L(M_2) = \{w \in \{a, b\}^* \mid w \text{ contains the substring } bb\}$

M_1 :



M_2 :



$$M_1 = (Q_1, \Sigma, \delta_1, q_0, F_1)$$

$$Q_1 = \{q_0, q_1\}$$

$$\Sigma = \{a, b\}$$

$$q_0 \in Q_1$$

$$F_1 = \{q_1\}$$

δ_1 is given as follows:

	a	b
q_0	q_1	q_0
q_1	q_0	q_1

$$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$$

$$Q_2 = \{q_2, q_3, q_4\}$$

$$\Sigma = \{a, b\}$$

$$q_2 \in Q_2$$

$$F_2 = \{q_4\}$$

δ_2 is given as follows:

	a	b
q_2	q_2	q_3
q_3	q_2	q_4
q_4	q_4	q_4

$$\text{Now } M = (Q, \Sigma, \delta, \{q_0, q_2\}, F)$$

$$Q = Q_1 \times Q_2$$

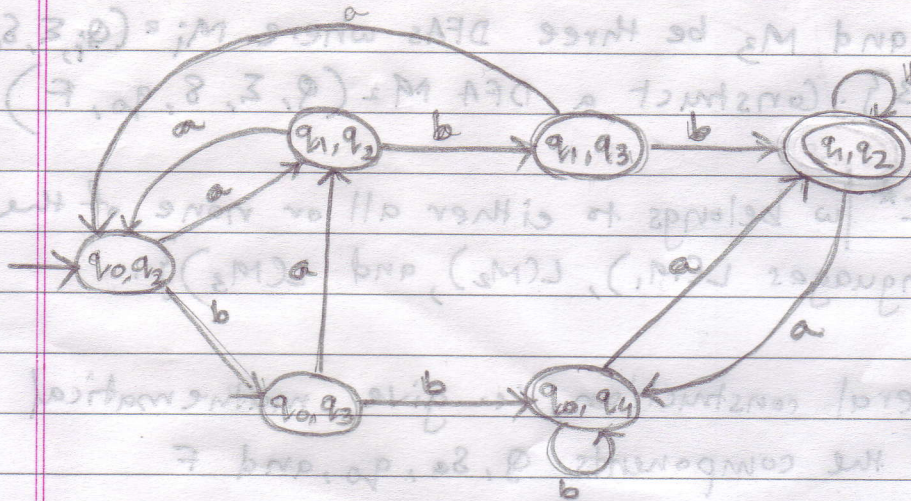
$$= \{\{q_0, q_2\}, \{q_0, q_3\}, \{q_0, q_4\}, \{q_1, q_2\}, \{q_1, q_3\}, \{q_1, q_4\}\}$$

$$\{q_0, q_2\} \in Q$$

$$\Sigma = \{a, b\}$$

$$F = F_1 \times F_2 = \{\{q_1, q_4\}\}$$

	a	b
$\{q_0, q_2\}$	$\{q_1, q_2\}$	$\{q_0, q_3\}$
$\{q_0, q_3\}$	$\{q_1, q_2\}$	$\{q_0, q_4\}$
$\{q_0, q_4\}$	$\{q_1, q_4\}$	$\{q_0, q_4\}$
$\{q_1, q_2\}$	$\{q_0, q_2\}$	$\{q_1, q_3\}$
$\{q_1, q_3\}$	$\{q_0, q_2\}$	$\{q_1, q_4\}$
$\{q_1, q_4\}$	$\{q_0, q_4\}$	$\{q_1, q_4\}$



Q.4 Consider the following language:

$$L_1 = (\{0\} \cup \{1\})^* \{0\} \{1\} (\{0\} \cup \{1\})^*$$

This language is a language of all strings over $\{0, 1\}$ that contain 01 as a substring. Notice that L_1 is expressed using the regular operations, and languages $\{0\}$ and $\{1\}$.

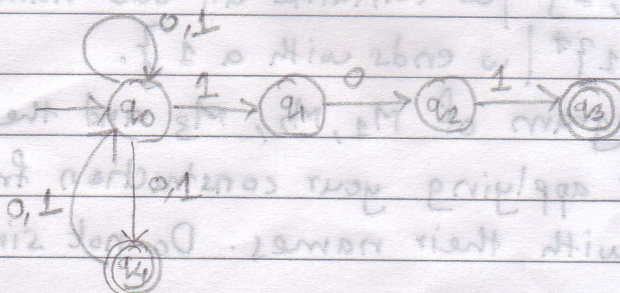
Let L_2 be the language of all strings over $\{0, 1\}$ that begin with the substring 101 or have exactly two 0's.

Express L_2 using the regular operations and the languages $\{0\}$, $\{1\}$, $\{e\}$ and $\emptyset$.

Soln

$$(\{1\}\{0\}\{1\}(\{0\} \cup \{1\})^*) \cup (\{1\}^* \{0\} \{1\}^* \{0\} \{1\}^*)$$

Q.5 Draw the state diagram of an NFA with at most seven states accepting the language of all strings over $\{0, 1\}$ that end in 101 or that have odd length.



Q.3. Let M_1, M_2 and M_3 be three DFAs where $M_i = (Q_i, \Sigma, \delta_i, q_i, R_i)$ for $i \in \{1, 2, 3\}$. Construct a DFA $M = (Q, \Sigma, \delta, q_0, F)$ such that

$$L(M) = \{w \in \Sigma^* \mid w \text{ belongs to either all or none of the languages } L(M_1), L(M_2), \text{ and } L(M_3)\}.$$

a) Give a general construction, i.e. give mathematical definitions of the components Q, δ, q_0 , and F

Soln - The machine M is given as

$$M = (Q, \Sigma, \delta, q_0, F)$$

such that

$$Q = \{(r_1, r_2, r_3) \mid r_1 \in Q_1, r_2 \in Q_2 \text{ and } r_3 \in Q_3\} = Q_1 \times Q_2 \times Q_3$$

$$F = \{(r_1, r_2, r_3) \mid (r_1 \in F_1, r_2 \in F_2 \text{ and } r_3 \in F_3) \text{ or } (r_1 \notin F_1, r_2 \notin F_2 \text{ and } r_3 \notin F_3) \text{ and } r_1, r_2, r_3 \in Q\}$$

$$q_0 = (q_1, q_2, q_3)$$

$$\delta((r_1, r_2, r_3), a) = (\delta_1(r_1, a), \delta_2(r_2, a), \delta_3(r_3, a))$$

where $a \in \Sigma$

b) Show how your construction works on the following concrete example: each of M_1, M_2 and M_3 is a two-state DFA; the automata accept the following languages:

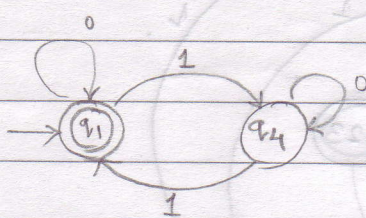
$$L(M_1) = \{w \in \{0, 1\}^* \mid w \text{ contains an even number of 1's}\}$$

$$L(M_2) = \{w \in \{0, 1\}^* \mid w \text{ contains an odd number of 0's}\}$$

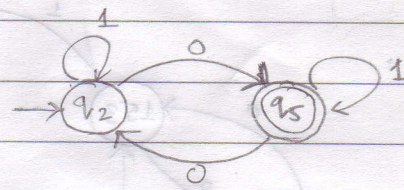
$$L(M_3) = \{w \in \{0, 1\}^* \mid w \text{ ends with a 1}\}.$$

Draw the state diagram of M_1, M_2, M_3 and the DFA M that is the result of applying your construction from part 3a. Label the states with their names. Do not simplify M .

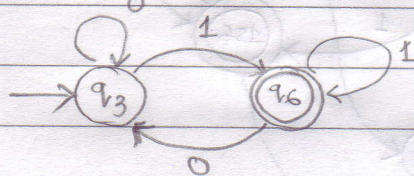
M_1 :



M_2 :



M_3 :



$$Q = \{ \{q_1, q_2, q_3\}, \{q_1, q_2, q_6\}, \{q_1, q_5, q_3\}, \{q_1, q_5, q_6\}, \\ \{q_4, q_2, q_3\}, \{q_4, q_2, q_6\}, \{q_4, q_5, q_3\}, \{q_4, q_5, q_6\} \}$$

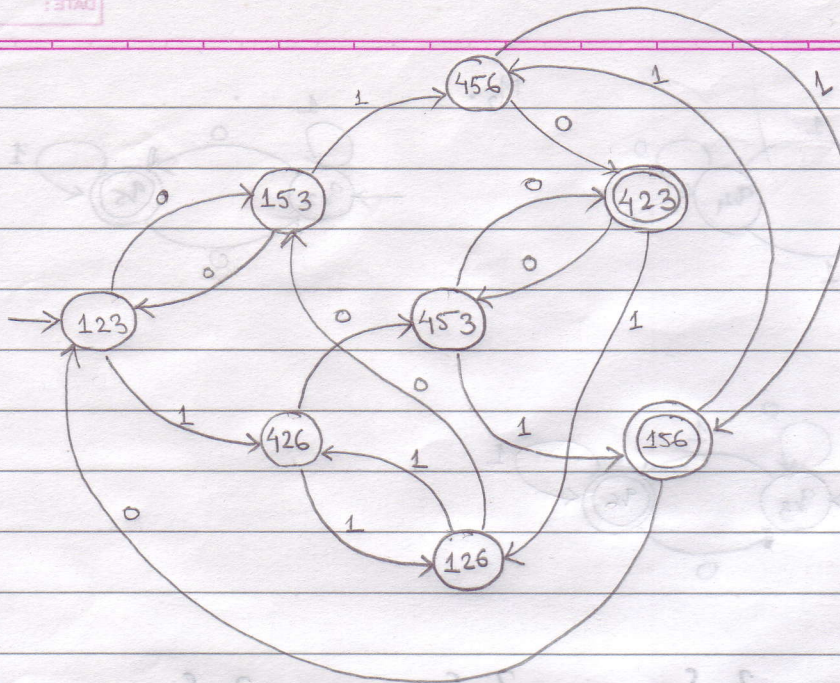
$$\Sigma = \{0, 1\}$$

$$F = \{ \{q_1, q_5, q_6\} \}$$

~~S is given~~ $q_0 = \{q_1, q_2, q_3\}$

S is given as

	0	1
$\{q_1, q_2, q_3\}$	$\{q_1, q_5, q_3\}$	$\{q_4, q_2, q_6\}$
$\{q_1, q_2, q_6\}$	$\{q_1, q_5, q_3\}$	$\{q_4, q_2, q_6\}$
$\{q_1, q_5, q_3\}$	$\{q_1, q_2, q_3\}$	$\{q_4, q_5, q_6\}$
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$\{q_4, q_5, q_6\}$	$\{q_4, q_2, q_3\}$	$\{q_1, q_5, q_6\}$



$$\{ \{a_p, a_q, a_r\}, \{a_p, a_q, a_r\}, \{a_p, a_q, a_r\}, \{a_p, a_q, a_r\} \} = 0$$

$$123 = \{q_1, q_2, q_3\}$$

$$153 = \{q_1, q_5, q_3\}$$

$$426 = \{q_4, q_2, q_6\}$$

$$456 = \{q_4, q_5, q_6\}$$

$$453 = \{q_4, q_5, q_3\}$$

$$126 = \{q_1, q_2, q_6\}$$

$$423 = \{q_4, q_2, q_3\}$$

$$156 = \{q_1, q_5, q_6\}$$

$$\{1, 0\} = 3$$

$$\{ \{a_p, a_q, a_r\} \} = 7$$

$$\{a_p, a_q, a_r\} = 0p \text{ move si } 2$$

$$0p \text{ move si } 2$$

1

0

$$\{a_p, a_q, a_r\}$$

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